

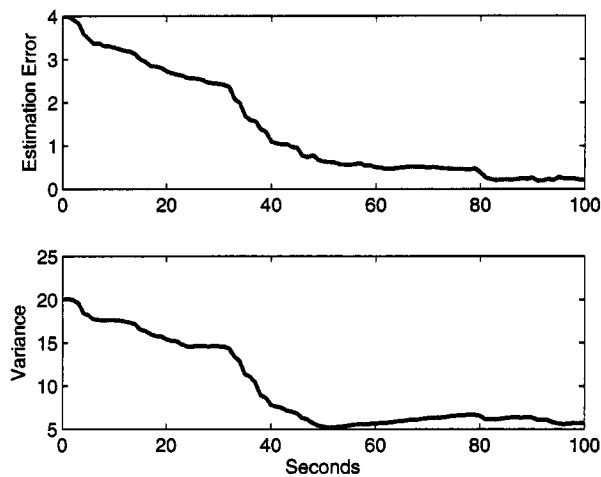


Figure 13.4 Example 13.4 results. Typical parameter estimation performance and parameter uncertainty for an extended Kalman filter estimating $-\omega_n^2$ for a second-order system. The estimation error of the unknown parameter and its variance gradually decrease toward zero.

to more readily adjust its estimate of $-\omega_n^2$, but also may prevent the filter from converging to the true value (see Problem 13.23).

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13.5 SUMMARY

Optimal state estimators can be derived for general classes of nonlinear systems as shown in [Kus67], but the filters are generally infinite dimensional, which makes them impractical for implementation. Finite-dimensional, optimal, nonlinear state estimators can be derived for more restricted classes of nonlinear systems [Liu80], but the restriction on the classes of applicable systems are significant enough to prevent wide applicability. Because of these factors, nonlinear Kalman filtering is the most widespread approach to state estimation for nonlinear systems.

It is interesting to note that the first applications of Kalman filtering were on nonlinear orbit-estimation problems [Bat62]. Some early investigations in nonlinear Kalman filtering can be found in [Cox64, Fri66]. Whereas stability and convergence results are readily available for the linear Kalman filter, such results are much more difficult to obtain for nonlinear Kalman filtering. Some convergence results for nonlinear Kalman filtering are found in [Urs80]. If the nonlinearities have known bounds then the Riccati equation can be modified in a simple way to guarantee stability for the continuous-time EKF [Rei98]. Conditions needed to guarantee the boundedness of the discrete-time EKF error covariance can be related to the observability of the underlying nonlinear system [Dez92, Son95].

PROBLEMS

Written exercises

13.1 Consider the scalar system

$$\begin{aligned}\dot{x} &= -x + w \\ y &= x + v\end{aligned}$$

The process noise has a mean value of 2, and the measurement noise has a mean value of 3. Redefine the noise quantities and the state to obtain an equivalent system of the form

$$\begin{aligned}\dot{x}' &= Ax' + Bu + w' \\ y &= Cx' + v'\end{aligned}$$

so that the new noise quantities w' and v' both have mean values of 0.

13.2 Consider the scalar system

$$\dot{x} = -x + u + w$$

w is zero-mean process noise with a variance of Q . The control has a mean value of u_0 , an uncertainty of 2 (one standard deviation), and is uncorrelated with w . Rewrite the system equations to obtain an equivalent system with a normalized control that is perfectly known. What is the variance of the new process noise term in the transformed system equation?

13.3 Suppose that x is a constant scalar, and $y_k = \sqrt{x}(1 + v_k)$ are noisy measurements, where $v_k \sim N(0, R)$.

- An intuitive way to estimate x is to set $\hat{x}_k = y_k^2$. Compute the mean and variance of the estimation error for this estimate. Your answer should be a function of x and R . Hint: recall that $E(v_k^3) = 0$ and $E(v_k^4) = 3R^2$.
- Perhaps a better estimate for x_k could be obtained by averaging all previous values of y_k^2 . That is,

$$\hat{x}_k = \frac{1}{k} \sum_{i=1}^k y_i^2$$

Compute the mean and variance of the estimation error for this estimate. Your answer should be a function of k , x , and R . Note that if you substitute $k = 1$ into your solution, you should get the same answer as part (a). What is the variance as $k \rightarrow \infty$?

- Write the extended Kalman filter equations to estimate x . What is the theoretical mean and variance of the EKF estimate as $k \rightarrow \infty$?

13.4 Consider the system

$$\begin{aligned}x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k^2\end{aligned}$$

where w_k and v_k are uniformly distributed, uncorrelated, zero-mean white noise processes with variances Q and R , respectively.

- a) What is the mean of the *a posteriori* estimation error for the discrete EKF?
- b) Modify the measurement equation by subtracting the known bias of the measurement noise so that the modified measurement noise is zero-mean. What is the variance of the modified measurement noise?

13.5 Consider the nonlinear system

$$\begin{aligned}x_{k+1} &= -x_k^2 + u_k + w_k \\ y_k &= 4x_k^2 + v_k\end{aligned}$$

Find the nominal values for x_k and y_k when $x_0 = 0$ and $u_k = 1$.

13.6 Consider the system $x_{k+1} = x_k^2 + w_k$, where w_k is zero-mean. The initial state x_0 is uniformly distributed between 0 and 1. An EKF is initialized with $\hat{x}_0^+ = E(x_0)$. What is $E(x_1)$? What is $\hat{x}_1^-$? This problem illustrates the fact that the state estimate of an EKF is not always equal to the expected value of the state.

13.7 Find the terminal velocity of the falling body of Example 13.2 if the terminal velocity occurs at an altitude of 1 mile.

13.8 Consider the hybrid scalar system

$$\begin{aligned}\dot{x} &= f(x) + w, & w &\sim N(0, Q) \\ y_k &= h(x_k) + v_k, & v_k &\sim N(0, R)\end{aligned}$$

The estimator that is used for the system is

$$\hat{x}_k = a + by_k + cy_k^2$$

Suppose that the state $x(t)$ is normally distributed with a mean of zero and a variance of P_x .

- a) Find an equation relating a , b , and c that must be satisfied in order for $\hat{x}_k$ to be an unbiased estimate of $x(t_k)$ [Gel74].
- b) Find values of a , b , and c so that $\hat{x}_k$ is the minimum-variance estimate. Assume that $h(x)$ is an odd function of x .

13.9 Suppose for a scalar system that $P_k^- = 1$, $R = 1$, and $H = 3$. What is the value of P_k^+ as given by Equation (5.19)? What will be the computed value of P_k^+ if $H = 2$ is used instead? What will be the computed value of P_k^+ if $H = 1$ is used instead? This illustrates how the iterated Kalman filter gets a more accurate estimate of P_k^+ by using a more accurate value for H_k .

13.10 Consider a system with the measurement equation $y_k = x_k^2 + v_k$. At time k the *a priori* state estimate is $\hat{x}_k^- = 1$, the true state is $x_k = 5$, and the measurement is $y_k = 25$. The *a priori* estimation-error variance is $P_k^- = 1$, and the measurement noise variance is $R_k = 4$. Use the iterated EKF algorithm to find $\hat{x}_{k,1}^+$ and $\hat{x}_{k,2}^+$. Although the iterated EKF does not always improve the *a posteriori* state estimate, this problem illustrates how it usually does.

13.11 Prove Lemma 6 for scalar random variables x .

13.12 Suppose you have the process equation $\dot{x} = x^2 + w$ and the state estimate $\hat{x}_k^+ = 0$. What is the differential equation for propagating $\hat{x}$ to the next measurement time using the first-order EKF? What is the differential equation using the second-order EKF?

13.13 Consider the measurement equation $y_k = x_k^2 + v_k$, where $v_k \sim (0, R)$. Suppose that $P_k^- = 1$, and $\hat{x}_k^- = 1$ is unbiased.

- What is the expected value of $\hat{x}_k^+$ if the first-order EKF is used for the measurement update? Based on your expression for $E(\hat{x}_k^+)$, how does the bias of the state estimate change with R ? Does this make intuitive sense?
- What is the expected value of $\hat{x}_k^+$ if the second-order EKF is used for the measurement update?

13.14 Consider the system

$$\begin{aligned} z_{k+1} &= az_k + w_k, & w_k &\sim (0, Q) \\ y_k &= z_k + v_k, & v_k &\sim (0, R) \end{aligned}$$

with unknown parameter a . Suppose that an EKF is used to estimate the state z_k and the parameter a . Further suppose that the artificial noise term used in the estimation of a is zero, and the EKF converges to the correct value of a with zero variance. Show that the EKF in this situation is equivalent to the standard Kalman filter for the scalar system when a is known.

Computer exercises

13.15 Write a program that implements the moving average filter and the extended Kalman filter for the system described in Problem 13.3. Use $R = 1$, $x = 1$, $P_0^+ = 1$, and $\hat{x}_0 = 2$. Which filter appears to perform better?

13.16 A planar model for a satellite orbiting around the earth can be modeled as

$$\begin{aligned} \ddot{r} &= r\dot{\theta}^2 - \frac{GM}{r^2} + w \\ \ddot{\theta} &= \frac{-2\dot{\theta}\dot{r}}{r} \end{aligned}$$

where r is the distance of the satellite from the center of the earth, θ is the angular position of the satellite in its orbit, $G = 6.6742 \times 10^{-11} \text{ m}^3/\text{kg/s}^2$ is the universal gravitational constant, $M = 5.98 \times 10^{24} \text{ kg}$ is the mass of the earth, and $w \sim (0, 10^{-6})$ is random noise due to space debris, atmospheric drag, outgassing, and so on.

- Write a state-space model for this system with $x_1 = r$, $x_2 = \dot{r}$, $x_3 = \theta$, and $x_4 = \dot{\theta}$.
- What must $\dot{\theta}$ be equal to in order for the orbit to have a constant radius when $w = 0$?
- Linearize the model around the point $r = r_0$, $\dot{r} = 0$, $\theta = \omega_0 T$, $\dot{\theta} = \omega_0$. What are the eigenvalues of the system matrix for the linearized system when $r_0 = 6.57 \times 10^6 \text{ m}$? What would you estimate to be the largest

integration step size that could be used to simulate the system? (Hint: recall that for a second-order transfer function with imaginary poles $\pm ja$, the time constant is equal to $1/a$.)

- d) Suppose that measurements of the satellite radius and angular position are obtained every minute, with error standard deviations of 100 meters and 0.1 radians, respectively. Simulate the linearized Kalman filter for three hours. Initialize the system with $x(0) = [r_0 \ 0 \ 0 \ 1.1\omega_0]$, $\hat{x}(0) = x(0)$, and $P(0) = \text{diag}(0, 0, 0, 0)$. Plot the radius estimation error as a function of time. Why is the performance so poor? How could you modify the linearized Kalman filter to get better performance?
- e) Implement an extended Kalman filter and plot the radius estimation error as a function of time. How does the performance compare with the linearized Kalman filter?

13.17 Implement the hybrid EKF with a measurement period of 0.1s for the system described in Example 13.1. Assume that the winding current measurement noises have a standard deviation of 0.1 amps. Create a table showing the experimental standard deviation of the motor velocity estimation error as a function of the standard deviation of the control input uncertainties q_1 and q_2 . Use control input standard deviations from 0 to 0.1 volts in steps of 0.01 (i.e., $\sigma_q = 0$, $\sigma_q = 0.01$, $\dots$, $\sigma_q = 0.1$). In order to make a fair comparison, you should either run several simulations for each value of σ_q and average the results, or else initialize the random seed in your software so that each simulation runs with the same random noise history.

13.18 Derive the first-order EKF, second-order EKF, and iterated EKF (with one iteration) for the scalar system

$$\begin{aligned}x_{k+1} &= x_k^2 + w_k \\ y_k &= x_k^2 + v_k\end{aligned}$$

where w_k and v_k are independent zero-mean white noise terms with variances 0.1 and 1, respectively. Simulate the first-order, second-order, and iterated extended Kalman filters for five time steps. Set the initial state to 1, the initial estimation-error variance to 1, and the initial state estimate to 2. Compute the RMS error of the filter estimates. How does the performance of the filters compare? (Note that you need more than one simulation, in general, to obtain a fair comparison of filter performance.)

13.19 Use the following procedure [Sor71b] to approximate a uniform pdf that is defined on ± 1 with M Gaussian pdfs; that is, $U(-1, 1) \approx \sum_{i=1}^M a_i N(\mu_i, \sigma_i^2)$.

- Select the weighting coefficients so that $a_i = 1/M$ for all i .
- Select the means of the Gaussian pdfs to be equally spaced on the range $[-1, 1]$ with $\mu_{i+1} - \mu_i = 2/(M+1)$.
- Select the variances σ_i of the Gaussian pdfs to all be the same and to minimize the RMS difference between $U(-1, 1)$ and $\sum_{i=1}^M a_i N(\mu_i, \sigma_i^2)$ over the range $[-1, 1]$.

The above approach reduces the approximation problem to a one-dimensional optimization problem, which can be solved in a number of different ways (for example,

using the golden search method [Pre92]). Plot the true pdf and the approximate pdf for $M = 3, 5$, and 10 , and compare the RMS errors.

13.20 Suppose you have a scalar system given as

$$\begin{aligned}x_{k+1} &= x_k \\ y_k &= x_k^2 + v_k\end{aligned}$$

where v_k is white Gaussian noise with a variance of 0.01 . The pdf of the initial state x_0 is uniform between -1 and $+1$. Note from the measurement equation that there is no way to distinguish between a positive state and a negative state.

- What will the extended Kalman filter estimate of the system be equal to?
- The pdf of x_0 can be approximated with two Gaussian pdfs, each with a variance of 0.43 , and with respective means of $-1/3$ and $+1/3$. Suppose that $x_0 = -1/2$. Plot the true state and the individual state estimates of a two-term Gaussian sum filter for 20 time steps. Plot the Gaussian pdfs at the final time for each estimate of the two-term Gaussian sum filter.

13.21 Consider the problem of tracking a moving vehicle in two dimensions (north is one dimension and east is the other dimension). The vehicle's acceleration in the north and east directions consists of independent white noise. Two tracking stations, located at north-east coordinates (N_1, E_1) and (N_2, E_2) , respectively, measure the range to the vehicle. The system model can therefore be written as

$$\begin{aligned}\begin{bmatrix} n_{k+1} \\ e_{k+1} \\ \dot{n}_{k+1} \\ \dot{e}_{k+1} \end{bmatrix} &= \begin{bmatrix} 1 & 0 & T & 0 \\ 0 & 1 & 0 & T \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_k \\ e_k \\ \dot{n}_k \\ \dot{e}_k \end{bmatrix} + w_k \\ y_k &= \begin{bmatrix} \sqrt{(n_k - N_1)^2 + (e_k - E_1)^2} \\ \sqrt{(n_k - N_2)^2 + (e_k - E_2)^2} \end{bmatrix} + v_k\end{aligned}$$

where n_k and e_k are the vehicle's north and east coordinates at time step k , T is the time step of the system, w_k is the zero-mean process noise, and v_k is the zero-mean measurement noise. Suppose that the time step $T = 0.1s$, the process noise covariance $Q = \text{diag}(0, 0, 4, 4)$, and the measurement noise covariance $R = \text{diag}(1, 1)$. The tracking stations are located at $(N_1, E_1) = (20, 0)$, and $(N_2, E_2) = (0, 20)$. The initial state of the vehicle $x_0 = [0 \ 0 \ 50 \ 50]^T$ and is perfectly known. Design an extended Kalman filter to estimate the state of the vehicle. Run the simulation for 60 s. Plot the estimation error for the four states. What is the experimental standard deviation of the estimation error for each of the four states? Based on the steady-state covariance matrix of the filter, what is the theoretical standard deviation of the estimation error for each of the four states?

13.22 Consider the system

$$\begin{aligned}x_{k+1} &= \phi x_k + w_k \\ y_k &= x_k\end{aligned}$$

where $w_k \sim (0, 1)$, and $\phi = 0.9$ is an unknown constant. Design an extended Kalman filter to estimate ϕ . Simulate the filter for 100 time steps with $x_0 = 1$,

$P_0 = I$, $\hat{x}_0 = 0$, and $\hat{\phi}_0 = 0$. Hand in your source code and a plot showing $\hat{\phi}$ as a function of time.

13.23 Simulate Example 13.4 with artificial parameter noise variance values $\sigma_p^2 = 0, 1$, and 100 . How does a change in the artificial parameter noise variance affect the filter's estimate of $-\omega_n^2$?

